LESSON 8 • TERM 1 • LEVEL 1

Balance and Swap

Addition and subtraction: the equals sign

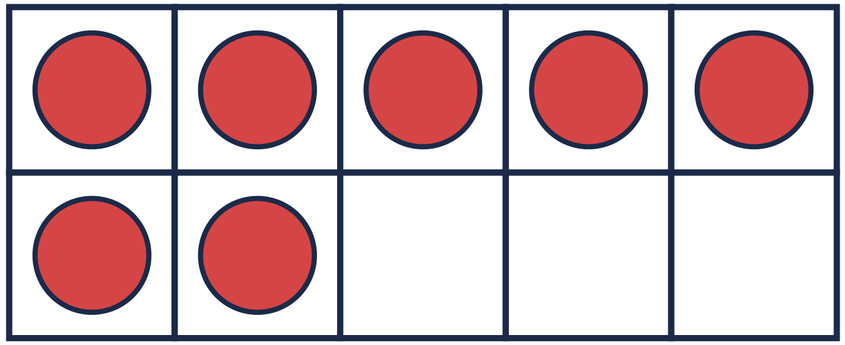
TODAY WE WILL

Balance two sides, and find missing numbers by giving and taking.



Warm-up: fingers to ten

2–8 MIN



I show 7. How many more fingers make 10?

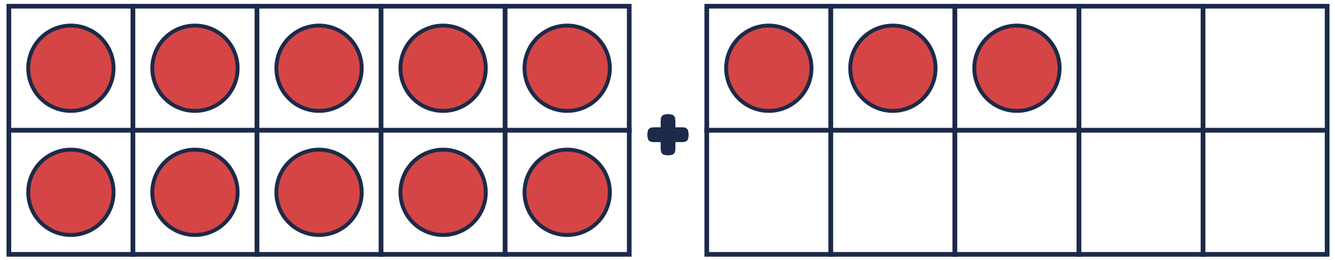
7 4 **9** 6 **2**

Show me your fingers!



Ten and more: how many to twenty?

2–8 MIN



Start at 13. How many more make 20?

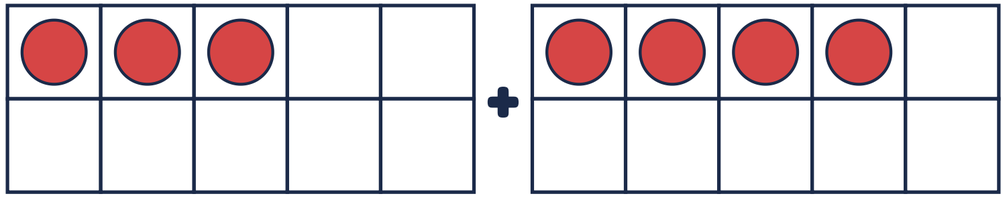
13 **16** **18** **15**

The equals sign

2–8 MIN

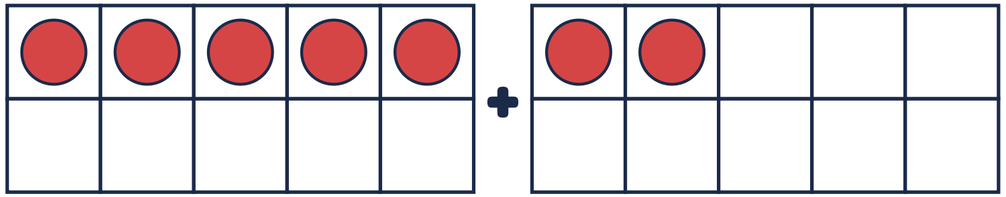
LEFT SIDE

3 + 4



RIGHT SIDE

5 + 2



= means the same amount as

Is 3 + 4 the same amount as 6 + 2?

Sneak-peek puzzles

8–12 MIN

Ideas we have not learnt yet. A guess is a great start. Nobody is marked.

PUZZLE 1: anyone can try

Ravi has 6 red beads and 4 blue beads.

Siti has 5 red beads and 5 blue beads.

Do they have the same number of beads? Draw dots to help, then circle Yes or No.

PUZZLE 2: think

8 + 5 = 9 + ___

What number goes in the blank? Write how you know.

Explore: two plates

12–28 MIN

1. Put your two plates side by side: LEFT and RIGHT.
2. Left plate: a pile of 6 and a pile of 4.
3. Right plate: a pile of 5, and a second pile.
4. Make the plates match. How many in the second pile?

WHAT IF?

What if the left plate has 7 and 3? What if it has 9 and 2?

12–28 MIN

Give and take

Left pile 7, right pile 5. Count all. Move 1 across. Count all again.

7 + 5 | 6 + 6 | **5 + 7** | 4 + 8

Next, match 8 + 4 with 9 + a second pile.

What did we notice?

12–28 MIN

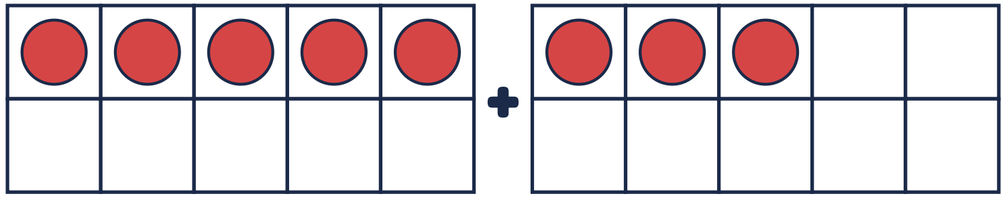
- When one number went up, what happened to the other?
- Did the total change when we gave one across?
- How could you find the second pile without counting everything?

Equals means balance

28–38 MIN

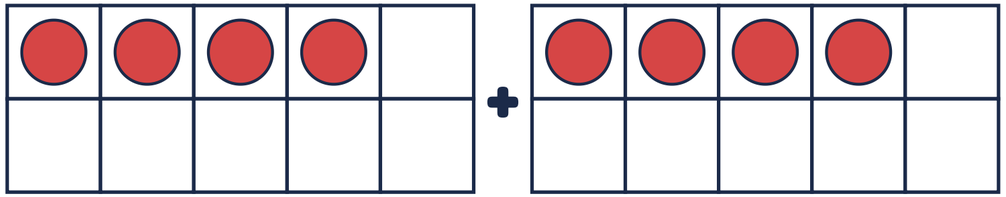
LEFT SIDE

5 + 3



RIGHT SIDE

4 + 4

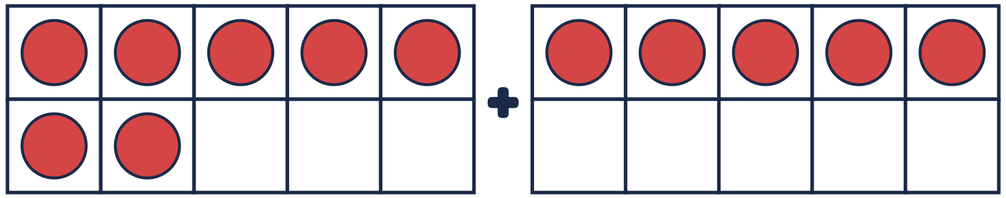


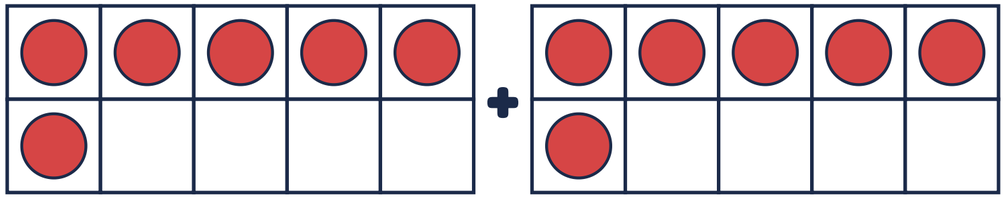
5 + 3 = 4 + 4. Both sides make 8.

Give and take

28–38 MIN

What we do	What we write
Start	7 + 5 = 12
Take 1 from the 7, give it to the 5	6 + 6 = 12
Again	5 + 7 = 12





The total stays the same.

28–38 MIN

Your turn: the missing number

Write on your paper. About 2 minutes

8 + 6 = 7 + ____

9 + 4 = 10 + ____

6 + 5 = 4 + ____

28–38 MIN

Check: the missing number

8 + 6 = 7 + 7

9 + 4 = 10 + 3

6 + 5 = 4 + 7

Bigger numbers, same idea

28–38 MIN

25 + 18 = 27 + ___

- 25 went up to 27: 2 more.
- So the blank is 2 less than 18.
- The blank is 16.

Check: 25 + 18 = 43 and 27 + 16 = 43.

Same total? Do not add yet

28–38 MIN

SAME TOTAL

47 + 18 and 45 + 20

47 goes down 2. 18 goes up 2.

NOT THE SAME

29 + 14 and 30 + 15

29 goes up 1. 14 goes up 1.

One number goes down and one goes up by the same amount. The total stays the same.

28–38 MIN

Quick-fire: is it balanced?

6 + 5 = 7 + 4

8 + 3 = 9 + 3

20 + 14 = 22 + 12

Thumbs up: balanced. Thumbs down: not balanced.

38–42 MIN

See-saw stand-up

Arms out wide like a see-saw. Left hand and right hand show the fingers.

6 and 4 | 5 and 5 | 7 and 3

Give one across. Still ten fingers! Sip some water.

Workbook page 1: Start here

42–70 MIN

Questions 1 to 3. About 8 minutes

S1

Fill in the missing numbers.

(a) 5 + 3 = 4 + ☐

(b) 6 + 4 = 5 + ☐

(c) 7 + 3 = ☐ + 7

(d) 2 + 8 = 8 + ☐

S2

Mei puts 6 buttons and 2 buttons on the left plate. On the right plate she puts 7 buttons and some more. Draw the right plate in the box so both plates are the same.

She adds ☐ more buttons

S3

Siti writes 6 + 3 = 5 + 9.

(a) Is it balanced? Circle Yes or No.

(b) Fill in the missing number: 6 + 3 = 5 + ☐

Check page 1

42–70 MIN

S1	4, 5, 3, 2
S2	1 more button (6 + 2 = 8 and 7 + 1 = 8)
S3	No. The blank should be 4 (6 + 3 = 9 and 5 + 4 = 9). Siti wrote the total of the left side, as if = meant the answer comes next.

Workbook page 2: Start here

42–70 MIN

Questions 4 to 7. About 9 minutes

S4

7 + 5 = 12. Fill in the missing numbers.

(a) 6 + ☐ = 12

(b) 5 + ☐ = 12

(c) 4 + ☐ = 12

S5

Fill in the missing numbers.

(a) 20 + 30 = 40 + ☐

(b) 50 + 20 = 60 + ☐

(c) 40 + 20 = 30 + ☐

(d) 30 + 30 = 20 + ☐

S6

Mei has 9 red pens and 6 blue pens. Ben has 10 red pens and some blue pens. They have the same number of pens. How many blue pens does Ben have?

Ben has ☐ blue pens.

S7

Fill in the missing number.

35 + 12 = ☐ + 2

Check page 2

42–70 MIN

S4	6, 7, 8
S5	10, 10, 30, 40
S6	5 blue pens (9 + 6 = 10 + 5)
S7	45 (35 + 12 = 47 and 45 + 2 = 47)

42–70 MIN

Choose your next page

Page	What it is
Think harder (page 3)	Same total without adding
Go further (page 4)	Two blanks and chains
Investigation (page 5)	Keep it the same
Check	Finish and check your own work

Everyone may choose. You can swap at any time.

Talk and explain

70–76 MIN

A CHILD SAYS

6 + 4 = 10. I add 1 to the 6 and 1 to the 4.

7 + 5 is still 10.

Do you agree? Use your plates to explain.

I think … because …

Back to the sneak-peek

70–76 MIN

8 + 5 = 9 + ___

The blank is 4. 9 is 1 more than 8, so the blank is 1 less than 5.

Exit ticket

76–82 MIN

On your own. No looking back. About 3 minutes

E1 **Fill in the missing number.**

9 + 5 = 10 + ▢

E2 **Is 28 + 6 = 30 + 4 balanced?**

Circle Yes or No.

E3 **Fill in the missing number.**

30 + 12 = 40 + ▢

Check the exit ticket

76–82 MIN

E1	4
E2	Yes
E3	2. The 30 went up by 10 to 40, so the 12 goes down by 10 to 2.

Look-back

82–85 MIN

Three minutes. There are no right answers. You can pass. Nobody reads it out.

Something I did well	write or draw one thing
The hardest part was …	☐ knowing what = means ☐ finding the blank ☐ how the numbers moved ☐ the big numbers (Think harder) ☐ something else
What helped me	☐ two plates and small things ☐ drawing it ☐ trying again ☐ I asked for help ☐ I was not stuck
Next time I will try	☐ drawing it ☐ using objects ☐ checking my answer ☐ asking a question
	☐ I am still stuck (I will ask in the class chat or show my tutor on camera) ☐ I would like to pass

Take your time. You can pass.



85–90 MIN

Well done, inquirers!

Show a grown-up one thing you found out today.

NEXT LESSON

Draw and Measure. Bring a cm ruler, 10 paper clips, a pencil and paper. No ruler? Tell your tutor.



OPTIONAL

Go further: what do you notice?

20 = 11 + ____

20 = 12 + ____

20 = 13 + ____

Keep going. What do you notice about the blanks?

OPTIONAL

If we have time: pick one

Activity	What we do
Is it always true?	Swap the two numbers in a sum. Does the total stay the same?
Make a puzzle	Write a missing-number puzzle for your grown-up
Teach it back	Show how to find the blank in 25 + 18 = 27 + ___

Optional. Everyone may join.